

Information Systems (Informationssysteme)

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Part III

A Very Brief Introduction to SQL

SQL—Structured Query Language

By far the most important query language today is **SQL**.

- Structured Query Language
- Originally meant to be used by end users ☺
- Today supported by virtually any database system

SQL operates on **relational data**:

Ingredients			
Name	Alcohol	InStock	Price
Orange Juice	0.0	12	2.99
Campari	25.0	5	12.95
Mineral Water	0.0	10	1.49
Bacardi	37.5	3	16.98

- Real databases may contain 100s or 1000s of **tables**, sometimes with billions of **rows** (also: **tuples**).

Our First SQL Query

The key construct of SQL is the SELECT-FROM-WHERE clause:

```
SELECT Name, Price  
      FROM Ingredients  
      WHERE Alcohol = 0
```

SELECT Choose a **set of columns** to be reported in the query result.



We'll later call this **projection**, **not** selection.

FROM Choose a **table** where rows should be taken from.

WHERE Additional **conditions** that rows must satisfy in order to appear in the result (the WHERE clause is optional).

→ **This** is what we call a **selection**.

Ingredients			
Name	Alcohol	InStock	Price
Orange Juice	0.0	12	2.99
Campari	25.0	5	12.95
Mineral Water	0.0	10	1.49
Bacardi	37.5	3	16.98

+

```
SELECT Name, Price
FROM Ingredients
WHERE Alcohol = 0
```

=

Name	Price
Mineral Water	1.49
Orange Juice	2.99

Data in Multiple Tables

Cocktail ingredients are sold by various suppliers (for a certain price), which could be represented as

SoldBy			
Ingredient	Supplier	DelTim ³	Price
Orange Juice	A&P Supermarket	1	2.49
Orange Juice	Shop Rite	3	2.79
Campari	Joe's Liquor Store	2	14.95
Bacardi	Liquor's & More	5	13.99
Mineral Water	Shop Rite	3	1.89
Bacardi	Joe's Liquor Store	2	14.99

³Delivery time in days.

When multiple tables are reference in the FROM clause, this is interpreted as the **Cartesian product** of the referenced tables:⁴

```
SELECT *
FROM Ingredients, SoldBy
```

Ingredients				SoldBy			
Name	Alcohol	InStock	Price	Ingredient	Supplier	DelTim	Price
Orange Juice	0.0	12	2.99	Orange Juice	A&P Supermarket	1	2.49
Orange Juice	0.0	12	2.99	Orange Juice	Shop Rite	3	2.79
Orange Juice	0.0	12	2.99	Campari	Joe's Liquor Store	2	14.95
Orange Juice	0.0	12	2.99	Bacardi	Liquors & More	5	13.99
Orange Juice	0.0	12	2.99	Mineral Water	Shop Rite	3	1.89
Orange Juice	0.0	12	2.99	Bacardi	Joe's Liquor Store	2	14.99
Campari	25.0	5	12.95	Orange Juice	A&P Supermarket	1	2.49
Campari	25.0	5	12.95	Orange Juice	Shop Rite	3	2.79
Campari	25.0	5	12.95	Campari	Joe's Liquor Store	2	14.95
Campari	25.0	5	12.95	Bacardi	Liquors & More	5	13.99
Campari	25.0	5	12.95	Mineral Water	Shop Rite	3	1.89
Campari	25.0	5	12.95	Bacardi	Joe's Liquor Store	2	14.99
Mineral Water	0.0	10	1.49	Orange Juice	A&P Supermarket	1	2.49
Mineral Water	0.0	10	1.49	Orange Juice	Shop Rite	3	2.79
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

⁴Use * in the SELECT clause when you simply want to choose all columns.

Queries over Multiple Tables

In practice, you rarely want to see this Cartesian product in the final result.

→ Use a WHERE clause to select only semantically related data.

```
SELECT Name, InStock, Supplier
FROM Ingredients, SoldBy
WHERE Name = Ingredient
```



Name	InStock	Supplier
Orange Juice	12	A&P Supermarket
Orange Juice	12	Shop Rite
Campari	5	Joe's Liquor Store
Mineral Water	10	Shop Rite
Bacardi	3	Liquors & More
Bacardi	3	Joe's Liquor Store

Queries over Multiple Tables

Resolve ambiguities by prepending column names with their table name:

```
SELECT Name, InStock, Supplier, SoldBy.Price  
FROM Ingredients, SoldBy  
WHERE Name = Ingredient  
AND SoldBy.Price < Ingredients.Price
```



Name	InStock	Supplier	Price
Orange Juice	12	A&P Supermarket	2.49
Orange Juice	12	Shop Rite	2.79
Bacardi	3	Liquors & More	13.99
Bacardi	3	Joe's Liquor Store	14.99

Tuple Variables

...or introduce **tuple variables** for easier reference:

```
SELECT Name, InStock, Supplier, s.Price  
FROM Ingredients AS i, SoldBy AS s  
WHERE Name = Ingredient  
AND s.Price < i.Price
```



Name	InStock	Supplier	Price
Orange Juice	12	A&P Supermarket	2.49
Orange Juice	12	Shop Rite	2.79
Bacardi	3	Liquors & More	13.99
Bacardi	3	Joe's Liquor Store	14.99

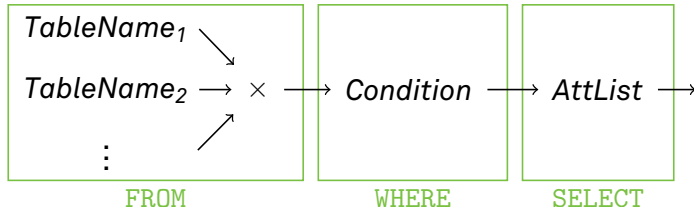
(The keyword AS is optional; 'SoldBy s' would mean just the same.)

Semantics of SQL SELECT-FROM-WHERE Expressions

Conceptually, the query

```
SELECT AttList
  FROM TableName1, TableName2, ...
 WHERE Condition
```

does the following:



(But most likely, the database system will choose a better strategy to actually execute the query.)

Concluding Remarks

- SQL is **case insensitive**; use ' as a **string delimiter**.
- It is okay to reference the **same table multiple times** in a FROM clause (→ “self-join”). Use **tuple variables** then to tell things apart.



Never, never ever, write queries where the correctness depends on the current table contents.

E.g., the correct answer to “give me names and prices of all non-alcoholic ingredients” is **not**

```
SELECT Name, Price
  FROM Ingredients
 WHERE Name = 'Orange Juice' OR Name = 'Mineral Water'
```